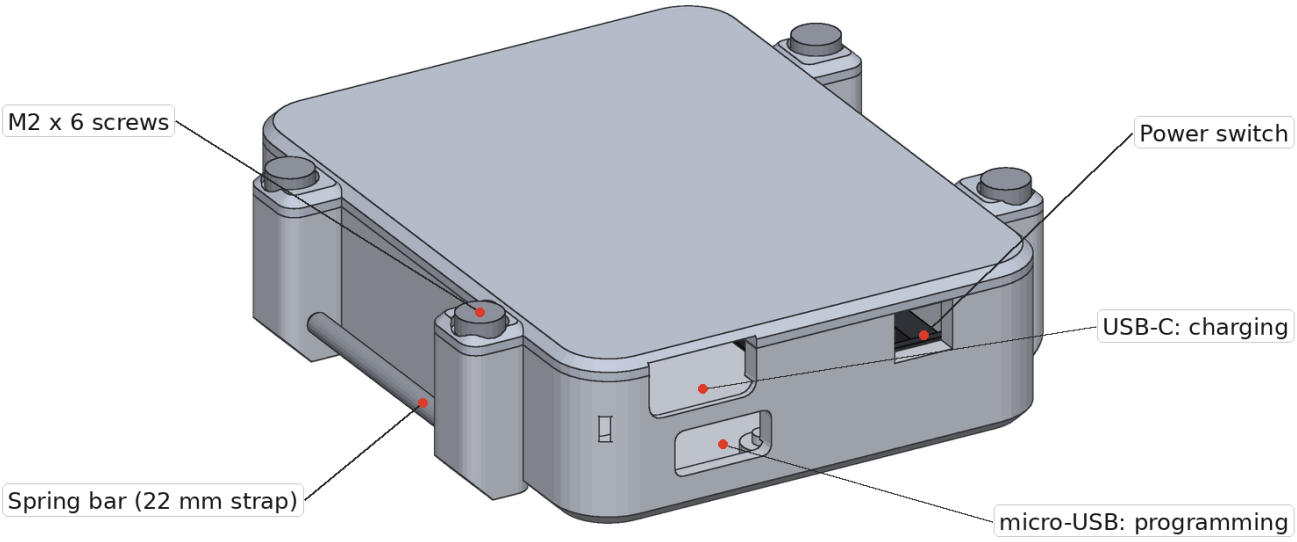
Sanad — wrist band body

3D-print files, wiring plan and step-by-step assembly: from the printed parts to wearing it



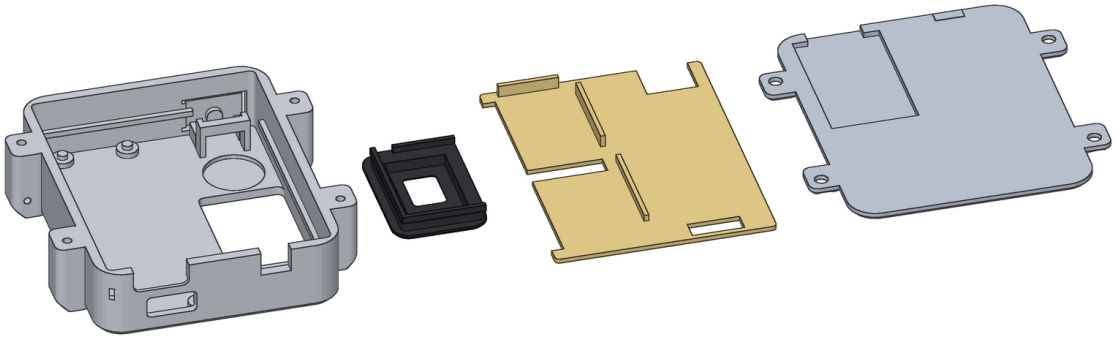
Body size	43 mm across the wrist × 54.6 mm along the forearm × 14.5 mm tall (+ 1 mm sensor bump underneath)
Strap	Any 22 mm quick-release watch strap (lug-to-lug 53 mm)
Printed parts	4: body, sensor pad, tray, lid — no supports needed
Inside	Raspberry Pi Pico 2 W, MAX30102, MPU-6050, TP4056 USB-C charger, 402030PT LiPo, push button, slide switch, coin vibration motor

Built from conservative estimates. Some sizes (the MAX30102 board, component heights and the parts you haven't bought yet) could not be measured, so the model uses safe upper-bound guesses: pockets are slightly roomy and get fixed with hot glue or foam tape. Page 4 lists every estimate and what it allows. If you can, check those numbers with calipers before printing; send me any number that's bigger than its allowance and I'll regenerate the files.

1. Ordering the print

You can send the files straight to an online 3D-printing service. STL and 3MF are the standard upload formats, so no conversion is needed. If the service asks for units, choose **millimetres**.

File	What it is	Qty	Sits on the bed	Colour
sanad-body.stl	Main case with lugs	1	bottom down (as worn)	any
sanad-sensor-pad.stl	Bump that holds the heart-rate sensor	1	bump down	**black**
sanad-tray.stl	Middle shelf between the two layers	1	flat, ribs up	any
sanad-lid.stl	Lid (already flipped for printing)	1	top face down	**white / natural**
sanad-band-all-parts.3mf	All four on one plate (same parts)	—	ready to print	



1 Body **2 Sensor pad** **3 Tray** **4 Lid (printed upside down)**

The four parts in their print orientation, rendered from the files.

Settings to ask for

Setting	Value	Why
Material	PETG (best) or PLA	PETG handles body heat and sweat and lets the button tab flex without cracking. Resin or nylon (MJF) also work well.
Layer height	0.12–0.16 mm	small holes and thin tabs come out cleaner
Walls / perimeters	4 (≈1.6 mm)	the side walls are 1.8 mm, so they print almost solid
Infill	40 %	
Supports	**None**	every part was designed to print without them
Lid colour	white or natural	the charger LEDs glow through a 0.6 mm thin window in the lid
Sensor-pad colour	black	keeps stray light out of the heart-rate sensor

2. What you need

Part	Notes	Qty
Raspberry Pi Pico 2 W	you have it (headers fitted)	1
MPU-6050 (GY-521)	you have it	1
MAX30102 module (green, 7 pins)	you have it	1
TP4056 Type-C charger with protection	Makers, 18 EGP. **Charge-current change needed — see Step 9**	1
LiPo 402030PT, 350 mAh real, with protection	Makers, 65 EGP	1
Mini push button 2-pin 6×6×5 mm	Makers, 1 EGP (buy 3)	1
SK12D07VG3 right-angle slide switch	Makers, 3.5 EGP	1
RCD1027 coin vibration motor (10 × 2.7 mm)	Makers, 15 EGP	1
1N5817 Schottky diode	battery → VSYS	1
2N2222A or S8050 NPN transistor (TO-92)	motor driver	1
1N4148 diode	across the motor	1
1 kΩ resistor	transistor base	1
4.7 kΩ resistor, 0603 or 0805 SMD	replaces the TP4056 charge-current resistor	1
M2 × 6 mm pan-head self-tapping screws	lid	4
22 mm quick-release watch strap	any brand	1
28–30 AWG silicone wire (4–5 colours)	about 1 m total	—
Kapton tape, 1 mm double-sided foam tape, heat-shrink		—

Tools

- Soldering iron with a fine tip, solder, flux, and desoldering wick for the TP4056 resistor swap
- Flush cutters, tweezers, small pliers, a hot-glue gun, and super glue (CA)
- Drill bits 1.2 mm and 1.6 mm in a pin vise, for cleaning holes by hand
- Small Phillips screwdriver, multimeter, and safety glasses — snipped pins fly

3. The estimated sizes

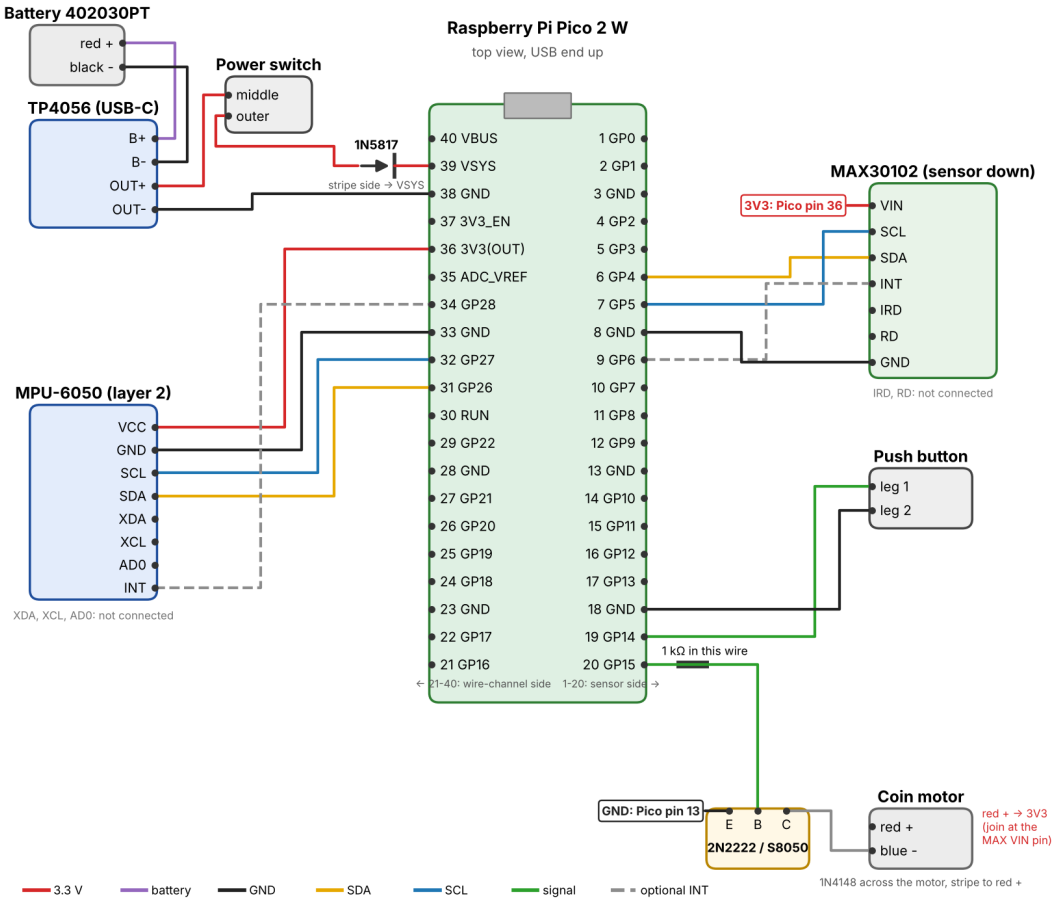
You asked me to go ahead without caliper measurements, so every number below is a **conservative guess**. Where possible it comes from the product photos (scaled by the 2.54 mm pin pitch, then perspective-corrected) or from typical part sizes. Pockets and openings are made at least this big, so parts up to these sizes fit. Smaller parts just sit a little loose and get held with glue or tape.

Value used	Size	Basis
Pico: tallest top-side part	3.2 mm	tallest part on the Pico top = micro-USB; typical micro-B receptacles are 2.6-3.0 mm, +margin
Pico micro-USB height	3.2 mm	micro-USB height above PCB; typical 2.6-3.0 mm, +margin
MPU-6050 PCB thickness	1.6 mm	standard 1.6 mm PCB
MPU-6050 tallest part	2 mm	tallest part (SOT-23 regulator ~1.1-1.45 mm), +margin
MAX30102 board length (pin-row direction)	17.9 mm	board length from the Makers back photo, scaled by the 2.54 mm pin pitch
MAX30102 board width	12.1 mm	board width from the same photo
→ pocket allows up to	19.2 mm	pocket allowance: photo estimate + 1.3 mm
→ pocket allows up to	13.4 mm	pocket allowance: photo estimate + 1.3 mm
MAX30102 PCB thickness	1.6 mm	standard 1.6 mm PCB
Sensor chip centre from the VIN edge	10.5 mm	chip centre from the VIN edge - perspective-corrected from the Makers front photo
Sensor chip centre from the pin-row edge	6.1 mm	chip centre from the pin-row edge - same photo analysis
Tallest other part on the sensor side	1.9 mm	yellow tantalum caps look like EIA 3216-18 (1.6 +/- 0.2 mm), +margin
TP4056 length	29 mm	Makers lists 28 x 17; another seller lists 29 x 17.3 x 4.3 - take the larger
TP4056 width	17.3 mm	larger of the two listings
TP4056 PCB thickness	1.6 mm	standard 1.6 mm PCB
TP4056 total height	5.2 mm	1.6 PCB + USB-C ~3.3; other listing says 4.3 total - take the larger
USB-C width	9 mm	USB-C receptacle shell is about 8.9 mm wide
USB-C height	3.3 mm	USB-C receptacle shell is about 3.2 mm tall
USB-C overhang	0.6 mm	board slides in a deep notch until the USB-C is flush, so this only sets the start position
Battery length incl. protection board	32.5 mm	30 mm cell + folded protection board and tabs
Battery width	20.5 mm	20 mm + tolerance
Button body	6.2 mm	6 x 6 mm listing + tolerance
Button plunger Ø	3.6 mm	typical 6x6 tact switch plunger 3.5 mm
Button plunger height	1.5 mm	5.0 total - typical 3.5 mm body
Switch knob width	2 mm	small slide-switch knobs are 1.5-2 mm
Switch knob length	3 mm	typical 2-3 mm
Switch travel	2.5 mm	typical 1.5-2 mm, +margin

Quick check worth doing: lay the MAX30102, TP4056 and battery on a ruler. If anything is **bigger** than its allowance (MAX30102 board 19.2 × 13.4 mm, TP4056 29.0 × 17.3 mm, battery 32.5 × 20.5 mm), tell me before you order the print.

4. Wiring plan

The two sensors are on separate I²C buses, which keeps every wire short. The MAX30102 uses I²C0 on the pins that face it, and the MPU-6050 uses I²C1 on the pins along the wire channel.



From	To (Pico pin)	Wire
Battery red / black	TP4056 B+ / B−	comes with the cell
TP4056 OUT+	Switch middle leg	≈30 mm
Switch outer leg	1N5817 end **without** the stripe	short, through the tray slot
1N5817 stripe end	39 VSYS	≈45 mm, across the Pico
TP4056 OUT−	38 GND	≈50 mm, through the TP4056 slot
MAX30102 VIN / GND	36 3V3(OUT) / 8 GND	≈40 / 25 mm
MAX30102 SDA / SCL / INT	6 GP4 / 7 GP5 / 9 GP6 (INT optional)	≈25 mm
MPU-6050 VCC / GND	36 3V3(OUT) / 33 GND	≈50 mm, down the wire channel
MPU-6050 SDA / SCL / INT	31 GP26 / 32 GP27 / 34 GP28 (INT optional)	≈50 mm, down the channel
Button legs	19 GP14 and 18 GND	≈25 mm
GP15 → 1 kΩ → transistor B	20 GP15	≈35 mm
Transistor E / C	13 GND / motor blue −	
Motor red +	MAX30102 VIN pin (3V3)	1N4148 across the motor, stripe to red +

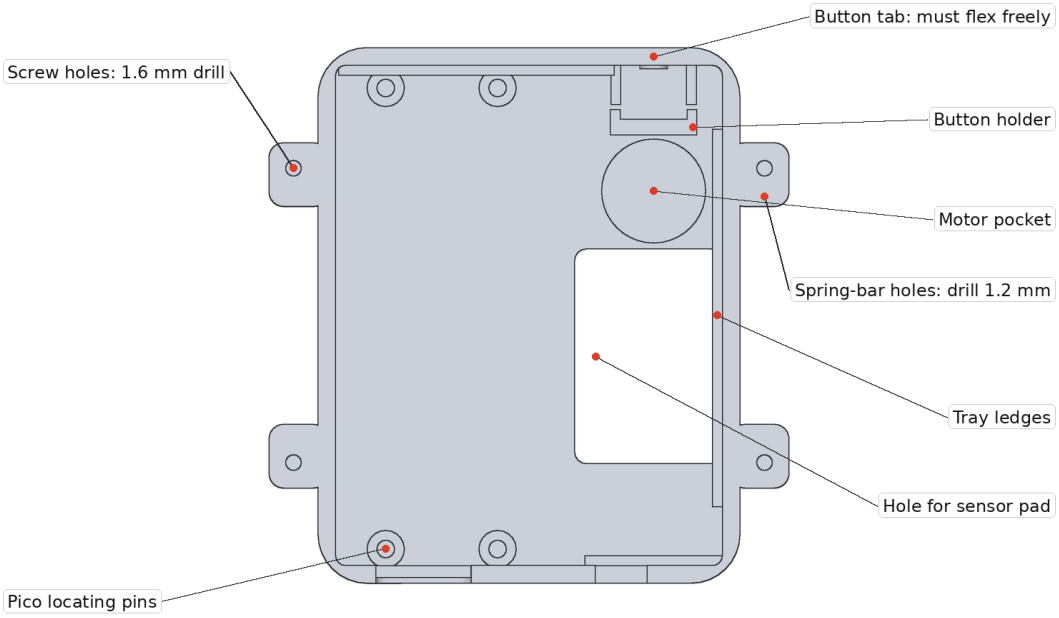
The battery reaches VSYS through a Schottky diode, as Raspberry Pi's Pico 2 W datasheet describes for a single lithium cell. The diode also stops the Pico's own USB from back-feeding the battery when you plug it in to program it. Charging only happens through the TP4056's USB-C.

5. Assembly, step by step

Work in this order: **the case first, then layer 1 on the floor, then the tray, then layer 2, then the lid**. Test before you close anything (Steps 6 and 9).

Step 1 — Clean up and check the print

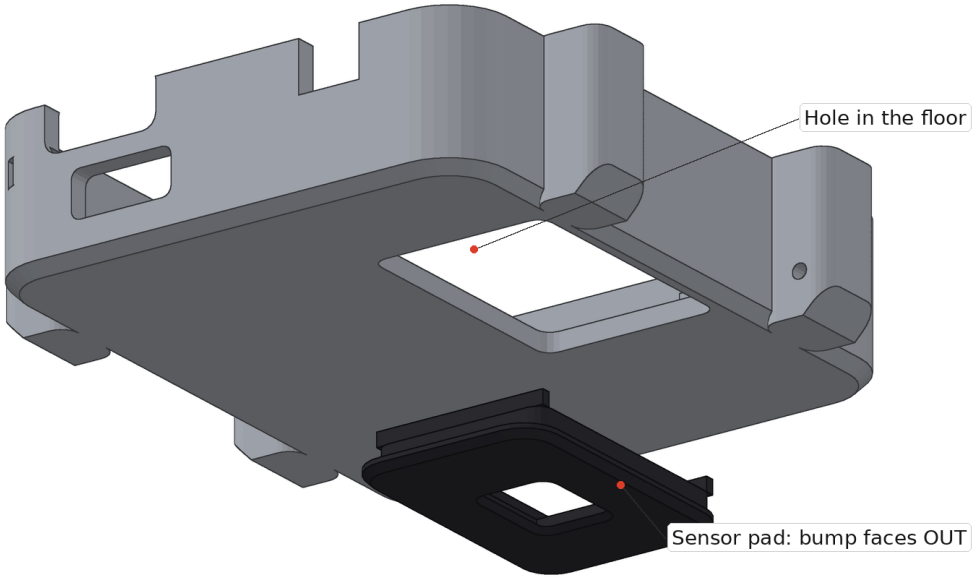
1. Remove any strings or brim. Run a **1.2 mm** drill by hand into the four spring-bar holes (on the inner faces of the lugs) and a **1.6 mm** drill into the four screw holes on top of the lugs.
2. Press the button tab on the side wall: it must flex a little and spring back. If a thin skin closed one of its U-slots, open it with a craft knife.
3. Check the four small locating pins on the floor are complete, and that the floor hole for the sensor pad is clean.



The body seen from above (button side at the top, ports side at the bottom).

Step 2 — Fit the sensor pad

1. Push the black sensor pad into the floor hole **from the outside**, bump facing out, until its rim sits flat against the bottom of the case. Its three low fences end up inside the case.
2. Fix it with a few drops of super glue around the plug, inside the case. Keep glue out of the window.



The pad goes in from below; the bump stays outside and presses against your skin.

Step 3 — Strip & snip the pin headers

1. Wear safety glasses. On each board, prise the black plastic strip up and off the pins: work a flat screwdriver under it, or nibble it apart with flush cutters.
2. Snip the pins. On the side that will get wires, leave a **1.5 mm** stub. On the other side, snip **flush**.
3. Re-touch every joint with the iron afterwards, because snipping can crack them.

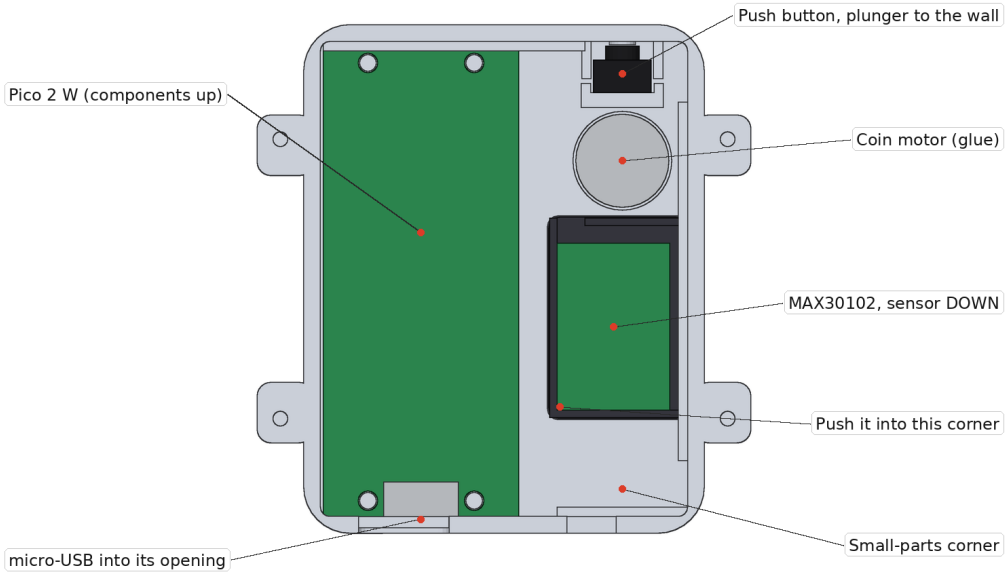
Board	Wire side (1.5 mm stubs)	Flush side
Pico 2 W	top (component side). Pins used: 6, 7, 8, 9, 13, 18, 19, 20 and 31, 32, 33, 34, 36, 38, 39	bottom (it rests on 1 mm pads)
MAX30102	back (the plain side)	**sensor side: must be flush**, it faces the pad
MPU-6050	bottom (the wires drop into the channel)	top

Step 4 — Pre-wire the boards

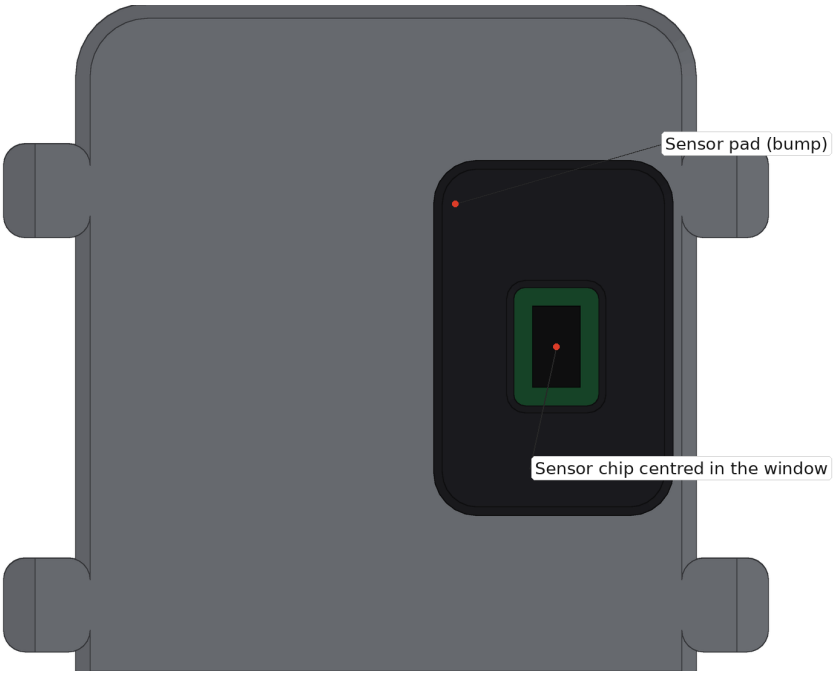
1. Solder wires to each board using the table on page 5. Cut each wire about 15 mm longer than the straight-line distance; you can shorten them during assembly.
2. Use one colour per signal: red 3V3, black GND, yellow SDA, blue SCL. Slide heat-shrink onto any splice before you solder it.

Step 5 — Layer 1: Pico, sensor, motor, button

1. **Pico 2 W**, components up: tilt it, lower the USB end first and slide the micro-USB into its opening on the ports side; the board's end slips into a slot under the opening. Then lower the far end onto the four locating pins.
2. **MAX30102**, **sensor face DOWN**, pin row toward the Pico, VIN end toward the ports side. Push it into that corner of the fences and check from underneath that the dark sensor chip sits in the window (figure below). Fix the board edges with hot glue; never on the chip.
3. **Coin motor** into its round pocket, wires up, held with a drop of hot glue.
4. **Push button** into its holder with the plunger against the wall, touching the nub on the button tab, and its two legs pointing left and right. Hot-glue it.



Layer 1 from above.



From underneath: the sensor chip must sit in the window. If it doesn't, slide the board until it does before the glue sets (the window has 1.3 mm of margin each way).

Step 6 — Small parts, then a first test

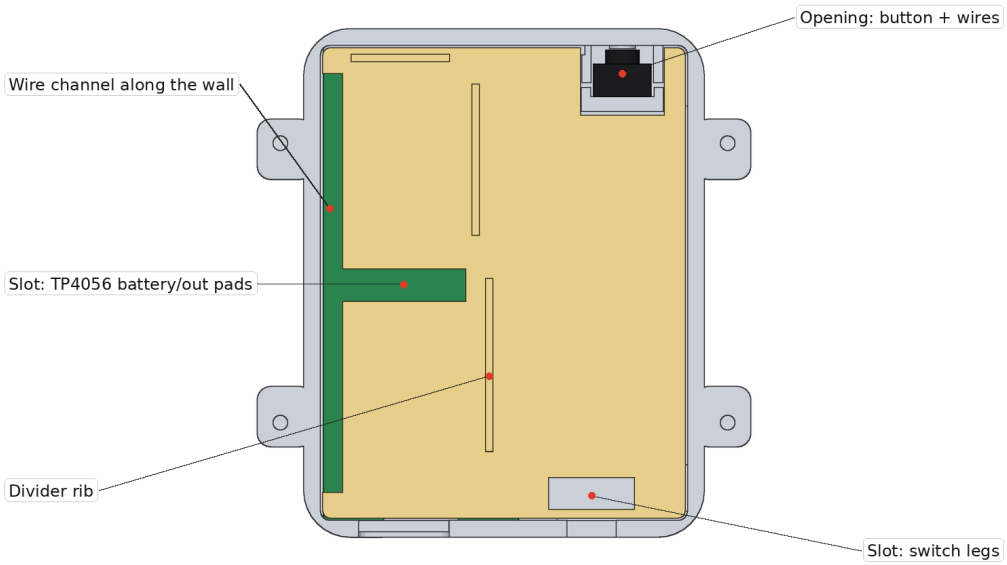
1. Put the 1N5817, the transistor with its 1 kΩ resistor, and the 1N4148 in the small-parts corner. Cover every bare lead with heat-shrink or Kapton.
2. Plug the Pico into your computer with micro-USB (battery not connected yet) and run this in MicroPython:

```
from machine import Pin, I2C
import time
i2c0 = I2C(0, sda=Pin(4), scl=Pin(5), freq=100000)    # MAX30102
i2c1 = I2C(1, sda=Pin(26), scl=Pin(27), freq=100000)  # MPU-6050
print('I2C0', [hex(a) for a in i2c0.scan()])   # expect ['0x57']
print('I2C1', [hex(a) for a in i2c1.scan()])   # expect ['0x68'] (needs the MPU wired)
motor = Pin(15, Pin.OUT); motor.value(1); time.sleep(0.4); motor.value(0)
btn = Pin(14, Pin.IN, Pin.PULL_UP)
while True:
    print('pressed' if btn.value() == 0 else '-'); time.sleep(0.3)
```

Press the button tab on the outside of the case. The output should change to **pressed**.

Step 7 — Tray

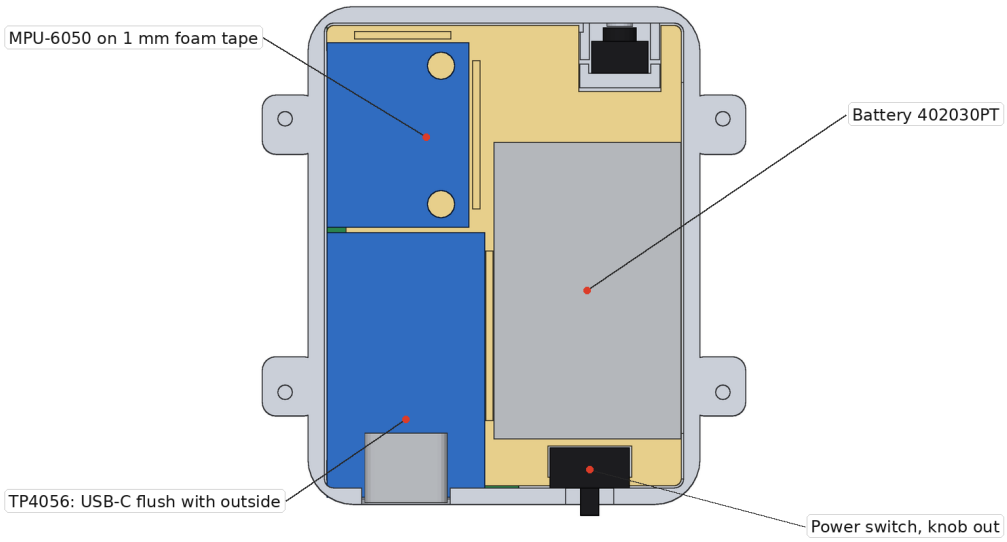
1. Pass the wires that go up to layer 2 through the tray: the MPU and power wires up the wall channel, TP4056 OUT− through the slot near the TP4056, and the switch legs' wires through the slot under the switch.
2. Lower the tray onto its ledges. The top of the button passes through the tray's opening.



Tray in place. Slots and channel are where the wires pass.

Step 8 — Layer 2: charger, battery, switch, MPU-6050

1. **TP4056**, components up, USB-C toward the ports wall: lower the USB end first into the opening, then slide it until the USB-C face is flush with the outside. A strip of Kapton under it, then hot glue.
2. **Battery** on 1 mm foam tape beside the divider rib, wires toward the TP4056 pads. Never pierce or bend the cell.
3. **Power switch**: knob out through its slot in the ports wall, legs down through the tray slot. Hot-glue it.
4. **MPU-6050** on 1 mm foam tape, header row over the wire channel and its wires going down it. Press it against its small rail.



Layer 2 from above.

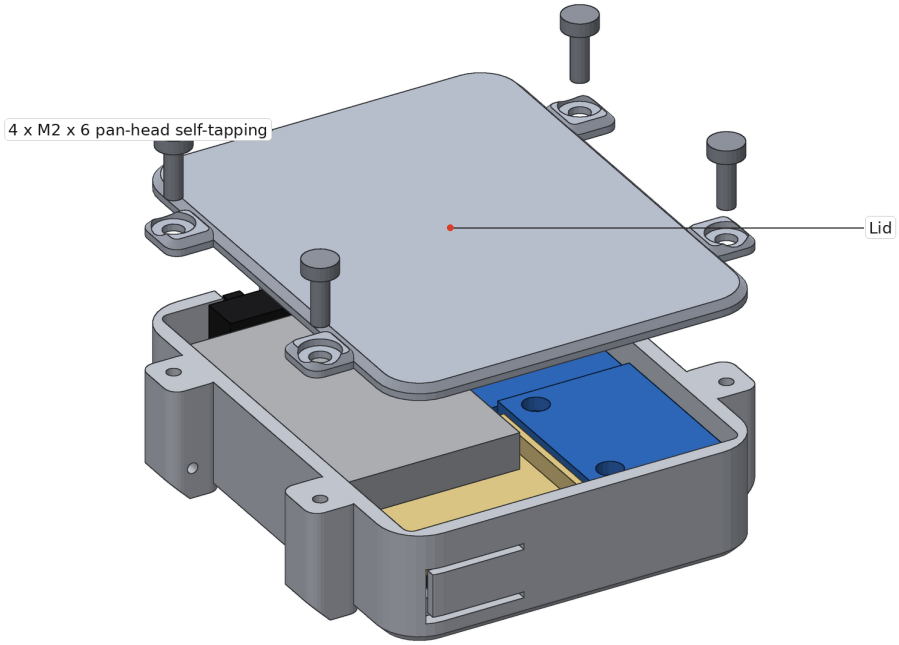
Step 9 — Charge current (important) and power test

Do this before the first charge. TP4056 boards usually ship set to about 1 A (1.2 kΩ, marked “122”). That is about 3× the capacity of a 350 mAh cell. The TP4056 datasheet table gives 300 mA at 4 kΩ and 250 mA at 5 kΩ. Replace the resistor next to the PROG pin (pin 2) with a **4.7 kΩ** one for about 250 mA. Until you've done it, don't charge this battery.

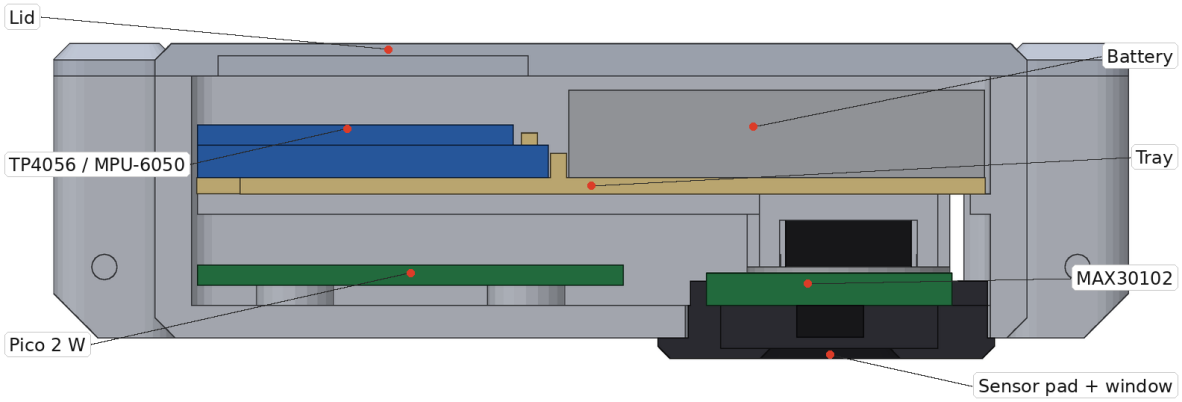
1. Solder the battery to B+ / B−. Switch the power switch ON. VSYS (pin 39) should read the battery voltage minus about 0.3 V, and the Pico should boot.
2. Plug a USB-C charger in. The red charging LED should glow through the lid's light window once the lid is on, and a different colour shows when the battery is full.
3. Switch OFF and check that VSYS drops to 0 V. The battery still charges with the switch off.

Step 10 — Close the lid

1. Tuck the wires flat so nothing presses on the battery. Place the lid: the thin light window goes over the TP4056, and the small tab fills the top of the switch slot.
2. Drive in the four M2 × 6 screws by hand until they're snug. Don't over-tighten; they cut their own thread in the plastic.



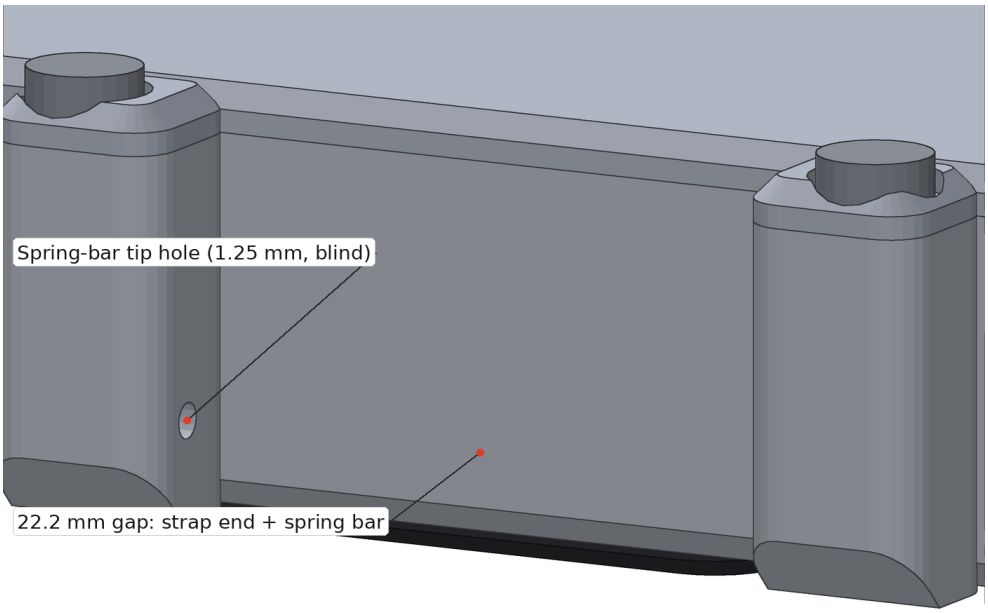
Lid and screws.



Cross-section through the sensor, cut from the real model. From the bottom up: sensor pad, MAX30102 and Pico on the floor, tray, then TP4056, MPU-6050 and battery, then the lid.

Step 11 — Fit the strap

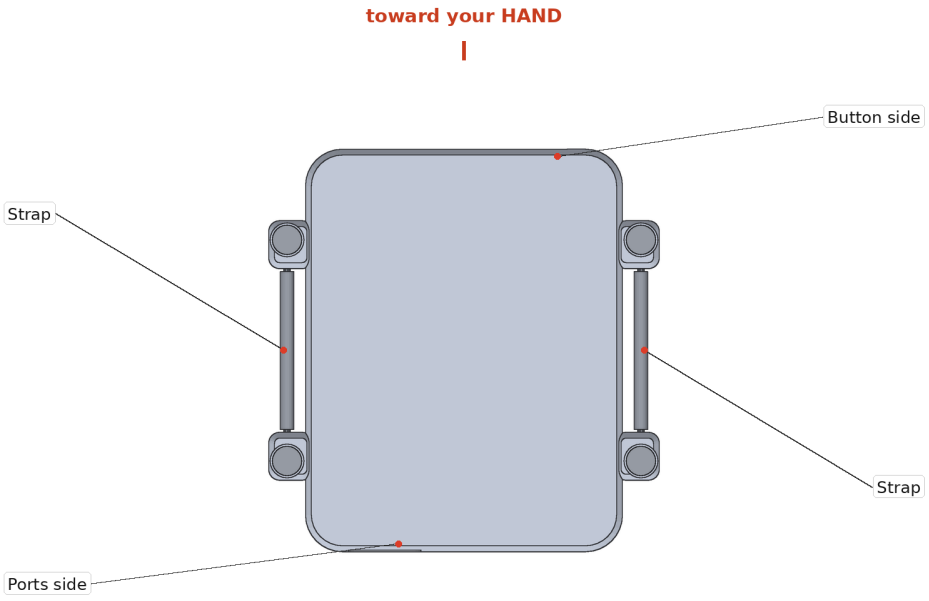
1. Slide a spring bar into the end of one strap half. Put one tip into a lug hole, pull the quick-release lever (or press the other tip with a pin) and let it snap into the opposite hole.
2. Tug it firmly. If a tip is loose, the hole printed too big: a drop of super glue inside the hole, left to dry, tightens it.



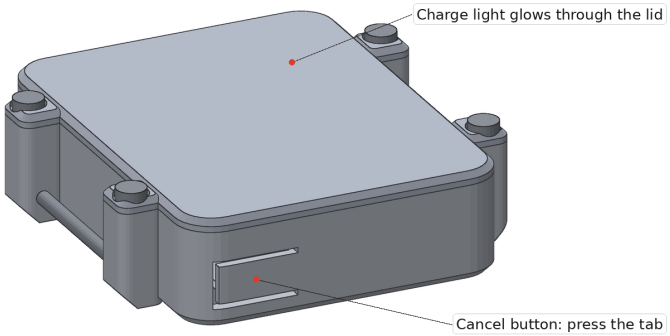
Spring-bar holes are blind, 1.25 mm, on the inner faces of the lugs; the strap end sits in the 22.2 mm gap.

Step 12 — Wear it

- Wear it on top of the wrist, just above the wrist bone. Point the **button side toward your hand** and the **ports toward your elbow**, so the button is where a watch crown would be.
- Tighten the strap until the sensor bump is pressed against the skin but you can still slide one finger under it. The heart-rate reading needs that contact.
- Charge through the USB-C on the ports side. You don't need to take the strap off to charge or to program.



Top view as worn.



Appendix A — Dimensions taken from real sources

Dimension	Value	Source
Pico length	51 mm	Raspberry Pi Pico 2 W datasheet (RP-008304-DS-3), Fig. 3 'The dimensions of the Pico 2 W'
Pico width	21 mm	Raspberry Pi Pico 2 W datasheet (RP-008304-DS-3), Fig. 3 'The dimensions of the Pico 2 W'
Pico PCB thickness	1 mm	Raspberry Pi Pico 2 W datasheet (RP-008304-DS-3), Fig. 3 'The dimensions of the Pico 2 W'
Pico hole Ø	2.1 mm	Raspberry Pi Pico 2 W datasheet (RP-008304-DS-3), Fig. 3 'The dimensions of the Pico 2 W'
Pico hole spacing (across)	11.4 mm	Raspberry Pi Pico 2 W datasheet (RP-008304-DS-3), Fig. 3 'The dimensions of the Pico 2 W'
Pico hole centre to end	2 mm	Raspberry Pi Pico 2 W datasheet (RP-008304-DS-3), Fig. 3 'The dimensions of the Pico 2 W'
Pico micro-USB width	8 mm	Raspberry Pi Pico 2 W datasheet (RP-008304-DS-3), Fig. 3 'The dimensions of the Pico 2 W'
micro-USB overhang	1.3 mm	Raspberry Pi Pico 2 W datasheet (RP-008304-DS-3), Fig. 3 'The dimensions of the Pico 2 W'
MPU-6050 length	20.26 mm	MPU6050 dimension drawing embedded on the Makers Electronics MPU-6050 product page (components101)
MPU-6050 width	15.56 mm	MPU6050 dimension drawing embedded on the Makers Electronics MPU-6050 product page (components101)
MPU-6050 hole Ø	3 mm	MPU6050 dimension drawing embedded on the Makers Electronics MPU-6050 product page (components101)
MAX30102 chip length	5.6 mm	Maxim MAX30102 datasheet p.1 (Yassen's upload): 14-pin OESIP optical module 5.6 x 3.3 x 1.55 mm
MAX30102 chip width	3.3 mm	Maxim MAX30102 datasheet p.1 (Yassen's upload): 14-pin OESIP optical module 5.6 x 3.3 x 1.55 mm
MAX30102 chip height	1.55 mm	Maxim MAX30102 datasheet p.1 (Yassen's upload): 14-pin OESIP optical module 5.6 x 3.3 x 1.55 mm
Battery thickness	4 mm	Makers Electronics listing 'LIPO Battery Cell 402030PT' (4.0 mm thick, 20 mm wide, 30 mm long)
Battery width	20 mm	Makers Electronics listing 'LIPO Battery Cell 402030PT' (4.0 mm thick, 20 mm wide, 30 mm long)
Battery length	30 mm	Makers Electronics listing 'LIPO Battery Cell 402030PT' (4.0 mm thick, 20 mm wide, 30 mm long)
Motor Ø	10 mm	Makers Electronics listing 'Mini Coin Vibration Motor RCD1027' (10 mm dia x 2.7 mm)
Motor thickness	2.7 mm	Makers Electronics listing 'Mini Coin Vibration Motor RCD1027' (10 mm dia x 2.7 mm)
Button total height	5 mm	Makers Electronics listing 'Mini Push Button Switch 2 Pin 6x6x5mm'
Switch length	8.7 mm	Makers Electronics listing 'SK12D07VG3 Slide Switch Right-Angle' (H 4.7, L 8.7, W 4.4)
Switch width	4.4 mm	Makers Electronics listing 'SK12D07VG3 Slide Switch Right-Angle' (H 4.7, L 8.7, W 4.4)
Switch height	4.7 mm	Makers Electronics listing 'SK12D07VG3 Slide Switch Right-Angle' (H 4.7, L 8.7, W 4.4)
Spring-bar tip Ø	0.9 mm	WatchGecko quick-release spring bars (0.8 mm tips, 1.8 mm body); Archer (tips ~0.9 mm)
Spring-bar body Ø	1.8 mm	WatchGecko quick-release spring bars (0.8 mm tips, 1.8 mm body); Archer (tips ~0.9 mm)
M2 pan head Ø	4 mm	ISO 7049 / ISO 7045 pan head, size 2: head dia 4.0 max, head height 1.6 max

Design choices: 1.8 mm walls, 1.6 mm floor and lid, 0.3 mm clearance around every board, 0.8 mm tray, 1.0 mm sensor bump, and 1.25 mm spring-bar holes (0.8–0.9 mm tips plus print tolerance).

Appendix B — Troubleshooting

Problem	Fix
A part doesn't fit its pocket	Don't force it. Send me the part's real size and I'll regenerate the STL in minutes; the model is fully parametric.
I²C scan doesn't show 0x57	Check SDA and SCL aren't swapped. Some MAX30102 boards pull the I²C lines up to 1.8 V only; if yours does, tell me and we'll add 4.7 kΩ pull-ups to 3V3.
Heart-rate signal weak	Tighten the strap a notch, make sure the chip is centred in the window, and raise the LED current in software.
Button doesn't click	The nub must touch the plunger. Add a 0.5 mm shim (tape) behind the button.
Lid won't close	Usually a wire under the battery. Re-route it through the wall channel.
Spring bar falls out	See Step 11, or use spring bars with longer tips.
Battery gets warm while charging	Unplug it and check the charge-current resistor (Step 9).